UNIVERSITY OF GHANA

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SECOND SEMESTER ASSESSMENT: 2026

BACHELOR OF SCIENCE IN BUSINESS ADMINISTRATION

(BUSINESS ANALYTICS OPTION)

MAIN CAMPUS

OMIS 308: PREDICTIVE ANALYTICS

ASSESSMENT FORM/SUBMISSION: GROUP/29 JULY 2026

Each group member is required to submit the report and Jupyter notebook individually via Sakai by 29 July 2026

INSTRUCTION:

ATTEMPT THIS QUESTION (20MARKS)

Case Study - *ED Happiness Estates*

Miss Osei Abrefi Akoto, CEO of *ED Happiness Estates*—a rapidly growing real estate firm in West Africa—has secured **$500 million USD** in funding from 35 venture capitalists and other global investors. The investment is intended for the development of **ultra-modern brutalist architecture buildings** for **rental and sale** in the world's busiest and most profitable cities.

Overwhelmed by investor expectations and uncertain about the most strategic locations, Miss Akoto has reached out to **your group (e.g., Team Google, Team Netflix, etc.)** to provide **data-driven advice** on which city or cities he should invest in.

Your team is required to use **real-world data** from multiple sources to generate **insights** into the factors that drive **housing and apartment prices/sales** in major global cities. You will then use these insights to advise Ms Akoto on the most suitable cities (or city) for investment.

To conduct this project, you are required to:

Choose **three major global cities** (e.g., Miami, Sydney, Accra, Cape Town, London, Berlin, Florida, Alberta, Tokyo, Delhi, Sao Paulo, Mumbai, New York and so on), at least one city from an advanced (developed) economy, one city from an emerging economy and one city from a developing economy.

(a) Provide a **clear justification** for selecting each city based on factors such as market size, growth potential, investor interest, economic stability, etc

(b) Collect real estate and socio-economic data for each city from **multiple reputable sources**, such as, Kaggle, World Bank Database, Numbeo, Zillow (for U.S. cities) and Government or

city-specific housing authorities. Ensure data relevance to housing prices, rental rates, population trends, income levels, and market demand.

(c) Clean and process the data, ensuring at least, **data integration, data transformation, outlier detection and handling, and feature engineering are performed before analysis.**

(d) Use python to conduct predictive analyses (or build predictive models) that explain determining factors of housing prices or sales. Ensure robustness of the results (evaluate model performance). Ensure that at least four predictive models are used/built/adopted, thus, either two **relevant** regression models and two **relevant** time-series models. Clear justifications should be provided for the choice of the models adopted for the analysis.

(e) Present and Interpret Your Results

(f) Based on your data analysis, findings and relevant accuracy checks, present reliable results to Ms Akoto. Also, provide **at least five (5) well-justified recommendations** for Ms Akoto. These should clearly indicate **which of the three selected cities** offers the best opportunity for real estate investment; the **rationale** behind each recommendation (e.g., high return on investment, market demand, population growth, rental yields, economic stability, etc.); support each recommendation with **data-driven insights** and relevant visualizations (e.g., comparison charts, price trends, risk analysis); **address potential risks or limitations** in each location and how they may impact the investment and **offer strategic guidance** on how *ED Happiness Estates* can position itself within each selected city's real estate market (e.g., targeting rental vs. sales, residential vs. commercial, luxury vs. affordable)

(g) **Report Formatting and Submission Guidelines**

Your final project report must adhere to the following requirements:

- Length: **15 to 30 typed A4 pages** (excluding references and appendices)
- Font: **Times New Roman**, font size **12**, with **1.5-line spacing**
- Structure: Include **a clear introduction, methodology, data sources, analysis, results, discussion, recommendations, and conclusion**
- Visuals: Use relevant **charts, graphs, tables, and figures** to support your analysis and enhance clarity
- Sources: Cite all data sources and references properly (following APA referencing style)
- Submission Platform: Upload the completed report and python file **individually** by each group member via **Sakai**
- Deadline: **29 July 2026**
- **Group Presentations (20-30 slides): To be confirmed later.** Presentation slides should not exceed 30 slides. The outline must align with the key themes outlined in the project brief, and the slides should be engaging and supported with relevant figures, tables, and chart.